

# International Macroeconomics of the Euro Area

## *A Modern, Model-Based Approach — Europe Focus, Interactive Edition*

Course Syllabus · Session 0 (toolkit) + 5 model sessions × 3h (21 teaching hours) · final exam scheduled separately · CY Cergy Paris Université

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Built on the slide decks for Schmitt-Grohé, Uribe & Woodford, *International Macroeconomics: A Modern Approach* (Princeton, 2022), publicly posted at [columbia.edu/~mu2166/UIM](https://columbia.edu/~mu2166/UIM). The model sequence is unchanged; US examples and cases are replaced with euro-area counterparts, and each session is anchored by a short video and a live data exercise students run themselves.

## Course Description

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This course follows the Uribe–Schmitt-Grohé UIM model of the open economy — a rigorous, model-based treatment built around a single intertemporal framework — and re-anchors it on the euro area. It covers, in order, the balance of payments and current-account sustainability, the intertemporal theory of the current account (endowment and production economies), the real exchange rate, international capital mobility, and monetary policy under alternative exchange-rate regimes, closing with currency pegs and balance-of-payments crises.

The distinctive European twist is that monetary union is treated as the limiting case of a hard fix: euro members share one interest rate and one currency, so the adjustment burden that a floating economy carries on its nominal exchange rate is redirected onto relative prices, wages, and employment. The euro periphery is the course's running laboratory.

The class combines 2nd- and 3rd-year undergraduates and does not assume a prior macroeconomics course. Session 0 supplies the five pieces of vocabulary the model sequence relies on: the national accounting identities, stocks versus flows, real versus nominal, discounting and present value, and exchange-rate conventions. It is taught in class and ends with a short graded check. The ungraded diagnostic at the start of Session 1 then measures whether it landed and tells the instructor which tool to revisit. Every model is built graphically first (no calculus required) and only then formalized algebraically, so the graphical route is the shared core and the algebra is the extension for stronger students.

## Interactive Design

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With one toolkit session and six model sessions of three hours each, every block is anchored by two active-learning elements: a short video (primary-source clip or neutral explainer) to open the mechanism, and a live data exercise where students pull real euro-area series themselves rather than seeing a static chart. All links are collected per session below and again in the session table.

- Videos are kept short (2–10 min) and neutral — primary sources (e.g. the Draghi speech) or mechanism explainers, not polemical documentaries — and are used to motivate a model, then dropped.
- Data exercises use free, maintained public portals (ECB Data Portal, Eurostat, the Osnabrück Euro Crisis Monitor) so students can reproduce every figure and extend it in a problem set.
- Each exercise has a 10-minute in-class 'core' version (pull one series, read it off the model) and an optional 'extension' (download the CSV, compute a ratio or spread) for the algebra track.

## Session-by-Session Plan with Media

### Session 0 — The Macro Toolkit (prerequisite session)

No chapter. Five tools the model sequence assumes, each with a worked numerical example and a live data exercise: the national accounting identities, ending at  $CA = S - I$ ; stocks versus flows and the NIIP accumulation relation; real versus nominal, including the Fisher relation and a first look at the real exchange rate; discounting, present value and the two-period budget constraint, with no-Ponzi and transversality stated in words; and reading an exchange-rate quote, closing on the observation that inside the union the nominal rate is frozen at one, so the whole adjustment has to run through prices and wages. The session ends with a graded toolkit check.

 **Video** [ECB Explainers hub](#) — short, neutral, primary-source clips. Assign “[What is inflation?](#)” before class as the warm-up for the real-versus-nominal block.

 **Data** Three 10-minute exercises, one per data-heavy tool. (1) [Eurostat nama\\_10\\_gdp](#): pull B1GQ, P3, P5G, P6, P7 for France and Spain and make the identity close, then compute  $S - I$ . (2) [Eurostat tipsii10](#) against [teibp010](#): find the year the flow turned and the year the stock turned, for Spain, Portugal, Greece and Germany. (3) [ECB inflation](#) and [ECB policy rates](#): compute an ex post real rate. Extensions download the CSV and plot the series.

 **Exchange rates** Read a live quote on the [ECB euro reference rates](#) page, state the convention out loud, then say which way the euro moved between two dates and what the same move looks like under the opposite convention. Extension: the Economist’s [Big Mac data](#) as the crudest possible real exchange rate, and the [ECB currency converter](#) to check the arithmetic.

✓ **Check** Toolkit check at the end of the session: ten short questions, two per tool, marked pass or fail, worth 5%. One retake before Session 2. The ungraded 15-minute diagnostic at the start of Session 1 repeats the same five questions and is used only to decide which tool to revisit.

### Session 1 — Global Imbalances, CA Sustainability & the Intertemporal CA

Chapters 1–3. BOP accounting, NIIP and sustainability, the two-period budget constraint and the endowment model. Framed on intra-euro imbalances, with TARGET2 balances as the visible financial-account counterpart to periphery current-account deficits. Opens with the ungraded diagnostic and a twenty-minute sketch of optimum-currency-area theory as the question the whole course answers; the formal treatment returns in Session 6.

 **Video** ECB explainer, “[What is the balance of payments?](#)” / [TARGET2 explainers](#) (ECB Explainers hub — short, neutral, primary source).

 **Data** Students open the [Euro Crisis Monitor \(Osnabrück University\)](#) and the [ECB Data Portal — TARGET balances](#), and watch the German claim rise as Greek/Spanish liabilities grow (2008–2012). Core: read the sign off  $S - I$ . Extension: download the CSV and line up TARGET2 against the CA.

### Session 2 — CA in a Production Economy & Fiscal Policy

Chapters 5 and 8. Investment/saving/CA schedules and adjustment to productivity shocks; twin deficits and Ricardian equivalence, applied to the Spanish and Irish convergence booms and the Stability & Growth Pact.

 **Video** Short explainer on the Spanish/Irish pre-2008 construction-and-credit boom (assign a neutral news-explainer clip; preview before class for tone).

 **Data** Students pull investment and current-account series for Spain and Ireland from [Eurostat \(national accounts / BOP\)](#) and locate the investment-driven CA deficit on the saving/investment diagram. Extension: compare to Germany over the same window.

### Session 3 — The Real Exchange Rate inside a Monetary Union

Chapters 9–10. PPP and its deviations, the TNT model, sudden stops. The conceptual pivot of the course: with no nominal exchange rate, 'real depreciation' means falling relative wages and prices — the periphery's 'internal devaluation'.

 **Video** Short explainer contrasting devaluation vs. internal devaluation (why Greece couldn't devalue but Iceland — outside the euro — did). Iceland is the deliberate counter-case.

 **Data** Students chart the ECB [Harmonised Competitiveness Indicators](#) for a periphery country: real appreciation to 2008, then post-2010 internal devaluation. Extension: decompose into the tradable/nontradable price gap.

### Session 4 — International Capital Mobility & the Euro

Chapters 11–12. Covered and uncovered interest parity; capital controls and their effects on consumption, investment and welfare. The forward-premium puzzle is flagged as an open research question rather than developed. Framed on euro-sovereign spread compression before 2008, the 2010–12 re-fragmentation, the sovereign-bank doom loop, and the Cyprus (2013) and Greece (2015) capital controls inside the union.

 **Video** Short news clip on the 2015 Greek bank closures / capital controls — capital mobility suspended inside a monetary union.

 **Data** Students pull 10-year sovereign yields for Germany, Italy, Spain, Greece from [ECB Data Portal — long-term interest rates](#) and plot spreads over the Bund: near-zero pre-2008, exploding 2010–12. Core: read it as (failed) interest parity. Extension: date the peak against 'whatever it takes'.

### Session 5 — Currency Pegs, Crises & the Euro Capstone

Chapters 14–15. Boom-bust under a peg, the anatomy of a BoP crisis, OCA theory, and the lender-of-last-resort backstop. Case studies: EMS 1992, the euro-area sovereign-debt crisis 2010–2012, and Greece as the periphery archetype.

 **Video** Draghi's 'whatever it takes' remarks, 26 July 2012 — primary source, hosted by the ECB: [verbatim remarks \(ECB\)](#). 90 seconds, then map it to the model as a lender-of-last-resort backstop that removes the run equilibrium.

 **Data** Students overlay the sovereign-spread series from Session 4 with the July 2012 date to see the break. Capstone: mixed-group case-study presentations; policy memo due.

 **Reading** Bianchi, Fang, Melosi & Rogantini Picco, [Why Europe needs Eurobonds \(VoxEU\)](#), 24 August 2026 — the union's missing fiscal leg. 2012 supplied the monetary backstop; the open question is whether stabilisation of a large common shock stays on national budgets or moves to a common, Eurobond-financed capacity, with national debt still nationally owed. Read the column only (the DSGE evidence stays out of scope) and discuss: with no exchange rate and one interest rate, which margin is left to adjust? Offered as a policy-memo prompt.

## Session Overview

Interactive anchors are in the right-hand column; the underlying model sequence is identical to the UIM course. Full links are in the session detail above.

| # | Block / Topic                                                        | Core content (European framing)                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 | The Macro Toolkit (prerequisite session)                             | National accounting identities, ending at $CA = S - I$ ; stocks versus flows and the NIIP accumulation relation; real versus nominal, the Fisher relation and a first look at the real exchange rate; discounting, present value and the two-period budget constraint, with no-Ponzi and transversality stated in words; exchange-rate conventions, appreciation versus devaluation, and the nominal rate frozen at one inside the union. Each tool taught with a worked numerical example. |
| 1 | Global Imbalances, CA Sustainability & the Intertemporal CA (Ch.1–3) | BOP accounting (current vs. financial account); CA, trade balance, NIIP; two-period intertemporal budget constraint; no-Ponzi & transversality; $CA = S - I = Y - A$ ; the endowment model and CA response to temporary vs. permanent shocks. Framed on intra-euro imbalances (German/Dutch surpluses vs. periphery deficits, 2000–08) with TARGET2 as the financial-account counterpart.                                                                                                   |
| 2 | CA in a Production Economy & Fiscal Policy (Ch.5, 8)                 | Investment schedule, saving schedule, CA schedule; equilibrium & adjustment to productivity shocks (graphical); twin-deficits intuition; Ricardian equivalence. Applied to convergence-driven investment booms in Spain and Ireland and the Stability & Growth Pact.                                                                                                                                                                                                                        |
| 3 | The Real Exchange Rate inside a Monetary Union (Ch.9–10)             | Law of one price, PPP and its deviations; nontradables; the TNT model — equilibrium & adjustment to shocks; sudden stops. Core euro theme: with no nominal exchange rate, real appreciation runs through relative prices/wages — the periphery's competitiveness loss and 'internal devaluation'.                                                                                                                                                                                           |
| 4 | International Capital Mobility & the Euro (Ch.11–12)                 | Covered & uncovered interest parity; capital controls — effects on consumption, investment, welfare. The forward-premium puzzle is flagged as an open research question, not developed. European framing: pre-crisis spread compression across euro sovereigns, the 2010–12 re-fragmentation, the sovereign-bank doom loop, and Cyprus (2013) / Greece (2015) capital controls inside the union.                                                                                            |
| 6 | Currency Pegs, Crises & the Euro Capstone (Ch.14–15)                 | Boom-bust cycles under a peg; anatomy of a balance-of-payments crisis; OCA theory; lender-of-last-resort backstop. Case studies: EMS 1992, the euro-area sovereign-debt crisis 2010–2012, Greece as the periphery archetype, and 'whatever it takes'.                                                                                                                                                                                                                                       |

## Assessment

| Component                  | Weight | Design notes                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Final exam (comprehensive) | 100%   | Separately scheduled, 3h. Covers the full model sequence plus euro-area policy applications. Graphical and intuitive throughout, with starred algebraic derivations carrying bonus credit only, so the two tracks are assessed on the final exactly as they are on the problem sets and the memo. A student who takes the graphical route the whole term is never required to derive. |

## Core Readings & Resources

- Schmitt-Grohé, S., Uribe, M., & Woodford, M. — *International Macroeconomics: A Modern Approach*, Princeton University Press, 2022 (primary text; Ch.1–5, 8–15).
- Uribe & Schmitt-Grohé UIM lecture slides by chapter — [columbia.edu/~mu2166/UIM](https://columbia.edu/~mu2166/UIM) — projected in class and for self-study.
- Obstfeld, M. & Rogoff, K. — *Foundations of International Macroeconomics*, MIT Press, 1996 (supplementary, formal treatment of intertemporal trade).
- Data portals for the in-class exercises: [ECB Data Portal](#), [Eurostat database](#), [Euro Crisis Monitor \(Osnabrück\)](#), [ECB Explainers](#).
- Bianchi, F., Fang, Q., Melosi, L. & Rogantini Picco, A. — “[Why Europe needs Eurobonds](#)”, VoxEU/CEPR, 24 August 2026 (short policy column, assigned in Session 6; the underlying two-country DSGE model is Bianchi, Fang, Melosi & Rogantini Picco, *Who is Afraid of Eurobonds?*, NBER WP 35510, 2026 — reference only, not assigned).
- European source packs (compiled, provided separately): the ERM/EMS crisis of 1992 (sterling & lira exit); the euro-area sovereign-debt crisis 2010–2012; and optimum-currency-area theory as the union backdrop.